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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations

December 2025

CV2105-1 – CONCRETE TECHNOLOGY

Duration: 3 hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Which of the following is not an ingredient of cement?
A) Lime
B) Silica
C) Sulphate
D) Magnesia
2. Progressive strength of OPC is contributed by presence of _____
A) C_3S
B) C_2S
C) C_4AF
D) C_3A
3. The compressive strength of cement is expressed as _____
A) kN/mm^2
B) kN/cm^2
C) N/mm^2
D) N/m^2
4. Which type of cement is suitable for mass concreting like construction of dams?
A) Expansive
B) high alumina
C) low heat
D) rapid hardening
5. Which of the following is not an air entraining admixture.
A) calcium chloride
B) aluminum powder
C) hydrogen peroxide
D) stearic acid
6. Which property of fine aggregates greatly influences the water demand of concrete?
A) Specific gravity
B) Grading
C) Absorption capacity
D) Bulk density
7. Which of the following impurities in water can lead to corrosion of steel in reinforced concrete?
A) Chlorides
B) Sulfates
C) Carbonates
D) Nitrates
8. BIS recommends that the permissible total dissolved solids (TDS) in water used for concrete should not exceed:
A) 2000 ppm
B) 3000 ppm
C) 5000 ppm
D) 6000 ppm
9. Which type of slump indicates a good workability and cohesive concrete mix?
A) Shear slump
B) Collapse slump
C) True slump
D) Zero slump
10. The measurement of workability by Vee Bee consistometer test is expressed in _____
A) minutes
B) centimeter
C) seconds
D) dimensionless
11. Which method of curing is suitable in cold regions?
A) electrical
B) chemical
C) high pressure steam
D) membrane

12. The height of the frustum of cone in slump test is _____
 A) 15 cm B) 30 cm
 C) 25 cm D) 50 cm
13. The rebound hammer test primarily provides information about which aspect of concrete?
 A) Flexural strength B) Surface hardness
 C) Compressive strength D) Workability
14. In the split tensile strength test, the specimen is usually in the form of a:
 A) Cylinder B) Cube
 C) Beam D) Slab
15. Which of the following chemical processes leads to the formation of ettringite in concrete?
 A) Chloride attack B) Carbonation
 C) Sulphate attack D) Freezing and thawing
16. Which property of concrete is critical for its long-term durability in an acidic environment?
 A) High water absorption B) High alkalinity
 C) Low porosity D) Low tensile strength
17. Sand of zone-I are:
 A) Coarse B) Fine
 C) Medium D) Very fine
18. What is the density of normal concrete?
 A) 1800 kg/m³ B) 2000 kg/m³
 C) 2600 kg/m³ D) 2400 kg/m³
19. Self-compacting concrete is designed to flow and fill formwork without the need for which of the following?
 A) Curing B) Vibration
 C) Reinforcement D) Water reduction
20. Which of the following is not a factor considered in the BIS method for selecting the mix proportions?
 A) Color of the cement B) Workability requirements
 C) Durability requirements D) Strength requirements

PART - B: DESCRIPTIVE ANSWER QUESTIONS

	Unit – I	Marks	BT*	CO*	PO*
1.	a) Explain the mixing process (wet process) involved in manufacturing of Cement with a neat flow diagram.	8	L*2	1	1
	b) Discuss the water quality parameters to check suitability of water for concrete.	8	L3	2	1,6,7
2.	a) Discuss the role of each Bogue's compound in the hydration process.	8	L3	1	1
	b) Explain the significance of grading of aggregates in concrete mix design.	8	L2	2	1
3.	a) Explain the standard consistency test of cement as per BIS and its significance.	8	L2	1	1
	b) Explain the role of fly ash in concrete, highlight the properties, advantages, and limitations.	8	L2	2	1
	Unit – II				
4.	a) Explain the procedure of the Vee-Bee consistometer test and its applications.	8	L2	3	1
	b) What is the gel/space ratio? Explain its influence on the strength of concrete.	8	L2	4	1

5.	a)	What is bleeding in concrete? Explain causes, effects, and remedial measures.	8	L2	3	1
	b)	Explain the method of determining the flexural strength of concrete.	8	L2	4	1
6.	a)	Explain the process of batching and transporting of concrete.	8	L2	3	1
	b)	What is carbonation of concrete? Explain its effects on reinforcement corrosion.	8	L2	4	1

Unit – III

7.	a)	Discuss the role of workability, strength, and durability requirements in mix design.	8	L3	5	1,6,7
	b)	What is High Volume Fly Ash Concrete? Explain its materials, properties, and applications.	8	L2	5	1,6,7
8.	a)	Explain the factors affecting concrete mix design.	8	L2	5	1,6,7
	b)	Discuss the importance of high-density concrete in radiation shielding.	8	L3	5	1,6,7

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
